

Biometric Authentication Mechanisms

Abstract—Biometric credentials are widely used in modern authentication: Sensors obtain a measurement and output *accept* if it is above a set match threshold or *reject* otherwise. A large body of work improves the accuracy of biometric sensors, but the state of the art uses a heuristic threshold. Other works study *authentication mechanisms*, but those treat biometric credentials as *atomic* (accept/reject, using the heuristic threshold).

We explore optimal authentication mechanisms with biometric credentials. Following prior work, we define a mechanism’s success as the probability it accepts legitimate users and rejects attackers. We formally prove that in general the optimal threshold differs from the common heuristic. Moreover, a different optimum should be used according to the integration with other credentials. We also prove that integrating a biometric credential improves mechanism success even when an atomic credential has arbitrary low fault probabilities and a biometric credential provides only weak separation between users and attackers. We quantitatively demonstrate the improvement with experiments on real public biometric data. Our optimized threshold reduces mechanism failure probability by 30-70% compared to the heuristic threshold when a biometric credential is integrated with other credentials.

These results imply that biometric sensors should allow integrators to set thresholds according to the authentication mechanism.

1. Introduction

Biometric credentials are one of the most prevalent keys in modern authentication. Historically, they were used in signatures and handprints on royal documents and contracts, as they rely on unique physical characteristics of the user made them unforgeable. In the early days of the Internet, biometric credentials were not applicable and therefore *atomic credentials*, like physical devices or passwords, either a perfect match or wrong, were adopted.

With the rapid growth of online services, more daily activities has transitioned to digital platforms. This expansion led companies to deploy advanced authentication mechanisms, such as multi-factor authentication (MFA) [1], increasing significantly the number of atomic credentials a user must have or remember to authenticate.

However, password vulnerabilities persist: 59% of users in the U.S. reported leaks and 22% frequently forget their passwords, as of 2026 [2]. Consequently, biometric credentials are increasingly adopted, present in most modern user devices, where access to most smartphones relies on FaceID or Fingerprint. The biometric sensors obtain a sample, cal-

culate a similarity score between a sample and a reference stored in the database, and output *accept* if the score is above a set match threshold or *reject* otherwise.

Previous work (§2) has conducted extensive analysis of biometric credentials in isolation, primarily focusing on improving the accuracy and latency, aiming to reduce the false acceptance (*FAR*) and false rejection (*FRR*) rates [3], [4]. Sarkar et al. [5] introduce the Equal Error Rate (*EER*, where $FAR = FRR$), a heuristic threshold of biometric sensors. Hong et al. [6] propose a fused biometric system in which the final decision is based on the matching scores of both face and fingerprint. While these works widely cover biometric credentials, none of them analyze their integration into an authentication mechanism. Other works analyze authentication mechanisms as predicates of credential possession [1], [7], [8], [9], where each credential is treated as atomic. However, such settings are partially applicable to our framework, as they treat biometric credentials as atomic. This leaves open the question of whether biometric credentials can be treated differently and if they can form the authentication mechanism when integrated with other credentials’ types.

In this work, we study how biometric credentials influence the authentication mechanism’s success with and without atomic credentials. More precisely, our goal is to optimally use biometric credentials in authentication mechanisms and analyze how the presence of atomic credentials may adapt the biometric credential threshold to enhance mechanism’s success. Authentication models (§3) are built on a set of biometric and atomic credentials. A mechanism interacts with two players: a user and an attacker, trying to identify the user based on the provided credentials by the player. Each credential can be either [9], (1) *safe*: Only available to the user, (2) *lost*: Not available to either player, (3) *leaked*: Available to both players, or (4) *stolen*: Only available to the attacker. Each credential has a probability of being in each of the four states. Biometric credentials are evaluated by a sensor that calculates a *confidence score* from an obtained sample. If the score is above a *match threshold*, the sensor outputs *accept*, otherwise *reject*. This is used to establish two fault probabilities: known as *false acceptance rate* (*FAR*) and *false rejection rate* (*FRR*).

We start by optimizing a single biometric credential mechanism (§4). We examine distribution functions that approximate score distribution as shown in Figure 1. We characterize each player by their score distribution and analyze several cases, reflecting whether the user and attacker are more likely to produce higher or lower matching scores s . We calculate the *FAR* and *FRR* metrics in each case

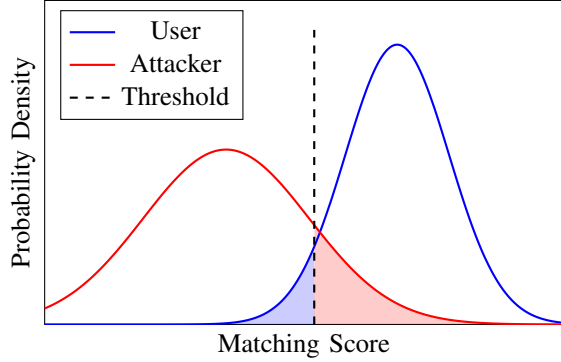


Figure 1: User and Attacker score distributions with *FAR* and *FRR* areas appear in red and blue respectively.

and derive closed-form expressions of the mechanism’s success. We then symbolically calculate the optimal and *EER* thresholds and show that in general, the optimal threshold differs from the commonly used *EER*. Moreover, we link the likelihood of an attacker to produce higher matching scores s to the gap between the maximum success and the success at *EER*. Our findings reveal that the more likely an attacker to produce higher matching scores s , the greater the benefit of using the optimal threshold instead of *EER*.

Having characterized the optimal threshold of a single biometric credential mechanism, we now extend our analysis to a more complex case, where a mechanism comprises a biometric and an atomic credential (§5). Such case is common in cloud authentication. For example, Google [10] assume they (1) Possess a trusted device like phone or computer (Atomic credential) and (2) Authenticates via Passkeys [11], e.g., FaceID or Fingerprint. Following the findings of Slobodan et al. [12] we examine first and second order approximations of a user and an attacker score distribution functions. We begin by analyzing a mechanism comprises two biometric credentials and evaluate its success using numerical optimization algorithm. Our results show that the optimal thresholds of the biometric credentials in such mechanism is not necessarily the same as the optimal threshold if used individually (§5.1). Moreover, to demonstrate the potential impact, we represent a synthetic experiment of two sampled score distributions and observe a reduction of over 66% in the failure rate, when comparing the *EER* and the optimal threshold. We then turn to analyze mechanisms that integrate both biometric and atomic credentials and derive closed-form expressions for the mechanism’s success. We symbolically calculate the optimal and *EER* thresholds and find the optimum depends on the values of the atomic credential. These expressions reveal that the mechanism success can primarily depend on either an atomic credential or a biometric credential, depending on the atomic credential probability values. For example, in an *AND mechanism*, the user has full availability to all credentials, while the attacker has limited or no availability. The derived relation shows that when the atomic credential has high safe probability and low leak probability, the mechanism’s success becomes

dominated by the atomic credential. Thus, changes in the biometric score, whether low or high, have only limited impact on the overall mechanism success; conversely, for other probability values of an atomic credential, success may depend more strongly on the biometric credential.

Motivated by this observation, we analyze whether the integration of biometric credential with an atomic credential will always enhance mechanism’s success. We link between the biometric credential to the improvement it has on the mechanism’s success composes an atomic credential and characterize the conditions under which such integration improves success (§6). We prove that when these conditions are not satisfied, it is not worthwhile to integrate, as the atomic credential mechanism dominates the integrated mechanism. Furthermore, we show that every biometric credential which is *asymptotic separable*: for every score above or below a specific threshold, the error rates converge to zero; a necessary and sufficient condition, provided that the atomic credential has non-zero probabilities for the states of *safe* and either *loss* or *leak*. We also introduce *asymptotic dominance*: where the genuine distribution dominates the attacker distribution by a factor c from every score above a specific threshold, or the attacker dominates the user by a factor c for every score below a specific threshold. This condition is a sufficient condition and shown as it represent a behavior we expect to observe in biometric credentials.

We quantitatively demonstrate our theoretical analysis with experiments using real public biometric data and a standard algorithm for fingerprint verification [13] (§7). We use a dataset of fingerprints from multiple users and evaluate matching scores by comparing samples of one user against those of other users. When different users attempt to authenticate as one another, the biometric sensors succeed in distinguishing and preventing unintended user access. In this setting, we observe high separability between the two players, as the algorithm effectively extracts minutiae and distinguishes between different individuals, resulting in low *EER*. Thus, no significant improvement is observed when comparing the mechanism’s success under Equal Error Rate (*EER*) threshold to the optimal threshold. However, since an attacker does not attempt to authenticate using their own biometric features, the relevant threat model involve attempts to physically impersonate a legitimate user. We conduct a spoofing attack experiment using the LivDet database [14], [15], [16], [17], which includes both genuine fingerprint scans and corresponding spoofing attempts. These spoof samples produce matching scores that closely resemble those of the legitimate user. Integrating the biometric credential into a mechanism comprised of an atomic credential, reduces the mechanism’s failure probability by 30-70% compared to the probability with the *EER* threshold. Furthermore, the results show a 50% reduction in the mechanism’s failure probability compared to a mechanism based solely on an atomic credential, demonstrating the benefit of integrating the biometric credential.

In summary (§8) our contributions are both theoretical and practical: (1) Formalization of the mechanism’s success function composed of biometric and atomic credentials;

(2) Symbolic analysis of the optimal threshold for biometric and atomic authentication mechanisms. (3) Characterization of the conditions under which biometric credential integration enhances the mechanism’s success; (4) Proof that any biometric credential, under a specified condition, always enhances the mechanism’s success. (5) Empirical analysis using real biometric data, demonstrating the theoretical analysis. (6) Direct implications to how fingerprint readers should work in practice for immediate security improvements.

2. Related Work

Authentication remains an active and prominent research area, with prior work broadly categorized into two main aspects: credentials and mechanisms. Credentials form the core of the authentication process, commonly manifested as passwords [18], OTPs [19], physical devices [20] or biometric features such as fingerprints, face recognition or iris [21].

Extensive work has evaluated and improved the accuracy of biometric systems [3], [14], [15], [16], [17], [22], [23]. Prabhakar et al. [4] examine the tradeoff between two key error metrics of a biometric system, namely False Acceptance Rate (*FAR*) and False Rejection Rate (*FRR*). Sarkar et al. [5] introduce the heuristic Equal Error Rate (*EER*), the point where *FAR* and *FRR* are equal and heuristically suggest it as an operating point of biometric credentials. However, none of these works define an optimizable success criterion for biometric systems, nor do they explore how the presence of additional credentials can influence the optimal matching threshold of the biometric system. Hong et al. [6] integrate fingerprint and face recognition to design a fused biometric system that combines the matching scores of both modalities to make a final decision. While their work is closely related to our domain, it doesn’t explore the influence of one biometric credential on the other, nor does it analyze the integration of biometric credentials along with other credential types within an authentication mechanism.

An authentication mechanism is a decision making protocol that evaluates a set of credentials provided by a player and output accept or reject [24], [25]. Examples include multi-factor authentication (MFA) [1], an authentication mechanism composes multiple and different credentials’ categories treating each one as an atomic credential. Recent work has explored authentication mechanism defined as predicate of credential possession. Maram et al. [7] study interactive authentication mechanisms, where the user and attacker interact with the authentication mechanism. Mouallem et al. [8] compare mechanisms’ success and aim to identify mechanisms with maximal success when composed of a large number of homogeneous credentials. Eyal [9] analyzes mechanisms modeled as Boolean functions of the credentials. However, these works are partially applicable to our setting. While we adopt their mechanism’s success definition and the way they define the credential possession states, they model biometric credentials as atomic

credentials. This leaves open the question of whether biometric credentials can be integrated with other credential types to form the authentication mechanisms with improved success, which is the focus of our work.

3. Model

We formalize the authentication problem by specifying the system participants (§3.1), Credential types and their characteristics (§3.2), the authentication mechanism (§3.3), and mechanism success (§3.4).

3.1. System Participants

The system comprises: an authentication mechanism M and two players: a user U and an attacker A . All three are finite and deterministic automata. Each player interacts with the mechanism M by sending and receiving messages. We abstract away the concrete implementation of the credential and model only the evidence that a player can present to the mechanism. Let:

$$\mathcal{C} = \{c_1, c_2, \dots, c_n\}$$

be the set of credentials used by the mechanism.

Definition 1. (Availability Vector) An availability vector represents the availability of all credentials to a user U or an attacker A . Denote by σ^U, σ^A the availability vectors of the user and the attacker, respectively.

A scenario σ is the vector of states of each credential in the mechanism. We denote by σ_i the state of credential c_i in scenario σ .

Before an authentication execution begins, the world fixes a scenario $\sigma = (\sigma^U, \sigma^A)$, where $\sigma^U, \sigma^A \in \{0, 1\}^n$.

For a player $P \in \{U, A\}$, the entry $\sigma_i^P = 1$ means that credential c_i is available to P , and $\sigma_i^P = 0$ means that it is not. We denote by

$$C_P(\sigma) = \{c_i \in \mathcal{C} : \sigma_i^P = 1\}$$

the set of credentials available to player P in scenario σ . A player can attempt to present evidence for any credential. For each credential c_i , we denote by $\text{present}_i(P, \sigma)$ the evidence produced by player P for credential c_i under scenario σ . If $c_i \notin C_P(\sigma)$, then the player cannot produce valid evidence for this credential, and we write

$$\text{present}_i(P, \sigma) = \perp.$$

Otherwise, the player produces an evidence value

$$e_i^P = \text{present}_i(P, \sigma).$$

Each player’s evidence is evaluated using a credential evaluation function:

$$y_i^P = \text{eval}_i(e_i^P).$$

Where the type of the evaluation function $\text{eval}_i(e_i^P)$ depends on the credential type.

3.2. Credentials

To authenticate, a player uses credentials to convince the mechanism that it is the legitimate user. A credential may correspond to secret data, such as a password or one time password (OTP); a physical object, such as a phone or security key; or a biometric property, such as a fingerprint or face.

Definition 2. (Atomic Credential) An atomic credential c^{atm} is evaluated as a binary outcome: either the presented evidence e_i^P perfectly matches the reference value, or the attempt is rejected as incorrect. For a player $P \in \{U, A\}$ and credential c^{atm} , we define

$$x_{c^{\text{atm}}}^P \in \{0, 1\},$$

where $x_{c^{\text{atm}}}^P = 1$ means that player P perfectly satisfies credential c^{atm} , and $x_{c^{\text{atm}}}^P = 0$ means that player P does not satisfy it. The probability of satisfying the credential is defined by $\Pr[x_{c^{\text{atm}}}^P]$ and the evaluation function is given by:

$$f_{c^{\text{atm}}}^P(x) = \Pr[x_{c^{\text{atm}}}^P = 1]\delta(x - 1) + \Pr[x_{c^{\text{atm}}}^P = 0]\delta(x),$$

where $\delta(\cdot)$ denotes the Dirac delta function.

Definition 3. (Biometric Credential) A biometric credential c^{bio} is evaluated by a biometric sensor. For a player $P \in \{U, A\}$ let s^P denote the *matching score* which quantifies the similarity between the input of P obtained by the sensor and the reference associated with c^{bio} . Let t denote the *match threshold*. If the score is above the threshold, the output of the biometric sensor is *accept*, otherwise *reject*. A biometric credential evaluation function is given by

$$f_{c^{\text{bio}}}^P(s) = \begin{cases} 1, & s_{c^{\text{bio}}}^P \geq t, \\ 0, & s_{c^{\text{bio}}}^P < t. \end{cases}$$

Each player uses a set of credentials $\{c_1, c_2, \dots, c_n\}$. Each credential c_i can be either *atomic* or *biometric* and can belong to one of the states: *Safe*, *Loss*, *Leak* or *Stolen*. We define the probability associated with each state as follows: P_{safe} , P_{loss} , P_{leak} and P_{stolen} . An attacker A and a user U have different availability to each credential. For example, a password may be available to the user but not to the attacker, while a leaked password may be available to both.

For example, if $\sigma_i = \text{leak}$ then the credential is available to both players, namely $\sigma_i^U = \text{True}$ and $\sigma_i^A = \text{True}$. If c_i is an atomic credential, we can deduce that $x_{c_i^{\text{atm}}}^U = 1$ and $x_{c_i^{\text{atm}}}^A = 1$. In the case where c_i is a biometric credential, this implies $f_{c_i^{\text{bio}}}^U(s) = 1$ and $f_{c_i^{\text{bio}}}^A(s) = 1$.

The biometric sensor compares the obtained sample with the enrolled reference template and calculates a matching score s . The genuine distribution $f_U(s)$ denotes the score distribution induced by samples produced by the legitimate user, whereas the impostor distribution $f_A(s)$ denotes the

score distribution induced by samples produced by the attacker. These distributions establish two error metrics: *FAR* and *FRR* both are functions of the match threshold t .

$$FAR(t) = \int_t^{a_2} f_A[s]ds$$

$$FRR(t) = \int_{u_1}^t f_U[s]ds$$

We use the *FAR* and *FRR* to find the probability of the biometric credential availability states. The user has access to the biometric credential when the output of the biometric sensor is *accept*, which occurs with probability $1 - FRR(t)$. The attacker has access to the biometric credential when the output of the biometric sensor is *accept*, which occurs with probability *FAR*(t). Thus, the probabilities of the four states are given by:

$$P_{\text{loss}} = FRR(t) \cdot (1 - FAR(t))$$

$$P_{\text{leak}} = FAR(t) \cdot (1 - FRR(t))$$

$$P_{\text{stolen}} = FRR(t) \cdot FAR(t)$$

$$P_{\text{safe}} = 1 - P_{\text{loss}} - P_{\text{leak}} - P_{\text{stolen}} = (1 - FAR(t)) \cdot (1 - FRR(t))$$

3.3. Mechanism

An authentication mechanism M interacts with two players: a user U and an attacker A . The mechanism receives the evaluation of each player's credential evidence and outputs a decision which is a function of the mechanism's logic.

Definition 4. (Authentication Mechanism) An authentication mechanism M is a finite deterministic function that takes as input:

- $C_P(\sigma)$ the set of credentials available to player P in scenario σ (perhaps no set, represented by $\perp$).
- σ_i the state of each credential c_i in scenario σ .
- y_i^P the evaluation of each credential c_i for player P (perhaps no evaluation, represented by $\perp$).

and outputs a variable decide_M , where $\text{decide}_M = 1$ means that the mechanism accepts the player and $\text{decide}_M = 0$ means that the mechanism rejects the player.

For example, in an *AND mechanism*, the mechanism grants access to a user if and only if all credentials are available to the user and at least one of them not available to the attacker. In an *OR mechanism*, the mechanism grants access to a user if at least one credential is available to the user and none are available to the attacker. We are now ready to define the mechanism's success.

3.4. Mechanism Success

A mechanism is successful in a scenario σ if the user is accepted and the attacker is rejected when the credentials available to them in σ are given as input. Formally,

Definition 5. (Mechanism Success) Mechanism success is the probability that the authentication mechanism M accepts a user U and rejects an attacker A given $C_P(\sigma)$, σ_i and y_i^P . For a player P the mechanism success is

$$P_{\text{Success}}(M) = \Pr[\text{decide}_M(U) = 1 \wedge \text{decide}_M(A) = 0].$$

4. Single Biometric Credential

The optimal threshold differs from the *EER*. We explore the change in optimal mechanism success probability as a function of the attacker's and user's variance. We start by analyzing single biometric credential mechanisms comprises first (§4.1) and second (§4.2) order approximations.

4.1. First-Order Approximation

We consider uniform distribution as a first-order approximation of a biometric credential's distribution. Denote by u_1 and u_2 the user's PDF bounds and by a_1 and a_2 the attacker's PDF bounds as shown in Figure 2. The probability distribution functions for the user and the attacker, f_U and f_A are thus:

$$f_U(s) = \frac{1}{u_2 - u_1} \quad \text{for } u_1 \leq s \leq u_2 \text{ and}$$

$$f_A(s) = \frac{1}{a_2 - a_1} \quad \text{for } a_1 \leq s \leq a_2.$$

The error metrics *FAR* and *FRR* are both functions of the match threshold t and given by:

$$FAR(t) = \int_t^{a_2} \frac{1}{a_2 - a_1} ds = \begin{cases} \frac{1}{a_2 - a_1} & \text{if } t \leq a_1 \\ \frac{a_2 - t}{a_2 - a_1} & \text{if } a_1 < t \leq a_2 \\ 0 & \text{if } t > a_2 \end{cases}$$

and

$$FRR(t) = \int_{u_1}^t \frac{1}{u_2 - u_1} ds = \begin{cases} 0 & \text{if } t \leq u_1 \\ \frac{t - u_1}{u_2 - u_1} & \text{if } u_1 < t \leq u_2 \\ 1 & \text{if } t > u_2 \end{cases}$$

Intuitively, genuine users are expected to produce higher matching scores than attackers. Therefore, the user score distribution is expected to be shifted toward higher matching scores s compared to the attacker distribution. Otherwise, the attacker would be more likely to produce scores that lead to successful authentication, rendering the biometric credential ineffective. Thus, we assume that $a_2 < u_2$. There are three distinct cases:

(1) $a_2 < u_1$, Figure 2a: In this case, the attacker and user distributions are separated. Any threshold $a_2 \leq t \leq u_1$ yields zero *FAR* and *FRR*, and therefore achieves maximal mechanism success probability.

(2) $u_1 < a_1 < a_2 < u_2$, Figure 2b: The attacker distribution bounds lies between the range of the user distribution bounds. For a mechanism M with a single credential c_1 , the mechanism's success probability is simply the probability that the credential is safe:

$$P_{\text{success}}(M^{\text{single}}) = P_{\text{safe}} \quad (1)$$

Since the credential is biometric, the success probability is:

$$P_{\text{success}}(t) = (1 - FAR(t)) \cdot (1 - FRR(t))$$

The *FAR* is one for any $t \leq a_1$ and zero for any $t \geq a_2$. Thus:

$$P_{\text{success}}(t) = \begin{cases} 0, & t \leq a_1 \\ (1 - \frac{a_2 - t}{a_2 - a_1}) \cdot (1 - \frac{t - u_1}{u_2 - u_1}), & a_1 < t \leq a_2 \\ 1 - \frac{t - u_1}{u_2 - u_1}, & a_2 < t \leq u_2 \\ 0, & t > u_2 \end{cases}$$

(3) $a_1 < u_1 < a_2 < u_2$, Figure 2c: The user and attacker distributions partially overlap.

We maintain same approach as the case where the attacker's support lies inside the user's support to calculate the probability of success. The *FAR* is one for any $t \leq a_1$ and zero for any $t \geq a_2$ while the *FRR* is zero for any $t \leq u_1$ and one for any $t \geq u_2$. Thus in this case we receive 4 intervals of the success probability function:

$$P_{\text{success}}(t) = \begin{cases} 0, & t \leq a_1 \\ 1 - \frac{a_2 - t}{a_2 - a_1}, & a_1 < t \leq u_1 \\ (1 - \frac{a_2 - t}{a_2 - a_1}) \cdot (1 - \frac{t - u_1}{u_2 - u_1}), & u_1 < t \leq a_2 \\ 1 - \frac{t - u_1}{u_2 - u_1}, & a_2 < t \leq u_2 \\ 0, & t > u_2 \end{cases}$$

We analyze the probability of success in both the case where the attacker's support lies inside the user's support and the overlapping case, and find the optimal threshold that maximizes the mechanism's success probability in each interval. The third and fourth regions of the overlapping case are equivalent to the second and third intervals in the case where the attacker's support lies inside the user's support, thus we analyze them together. First, we show that $P_{\text{success}}(t)$ is continuous. Since $P_{\text{success}}(t)$ is continuous on each interval of its piecewise definition, it remains only to check the boundary points. At each boundary point, the left-hand and right-hand limits of P_{success} coincide with P_{success} itself; therefore, P_{success} is continuous on its entire domain. In the overlapping case, the success probability function is linearly increasing in $a_1 \leq t \leq u_1$, and therefore its maximum is achieved at the right boundary, $t = u_1$. In both cases, the success probability is linearly decreasing on $a_2 \leq t \leq u_2$, and therefore its maximum is achieved at the left boundary, $t = a_2$. Since $t = u_1$ and $t = a_2$ are the boundary points of the interval where *FAR* and *FRR* are strictly between zero and one, the maxima of the adjacent regions are already included in this interval. Therefore, it is sufficient to analyze the interval where $u_1 \leq t \leq a_2$ in the overlapping case or $a_1 \leq t \leq a_2$ in the case

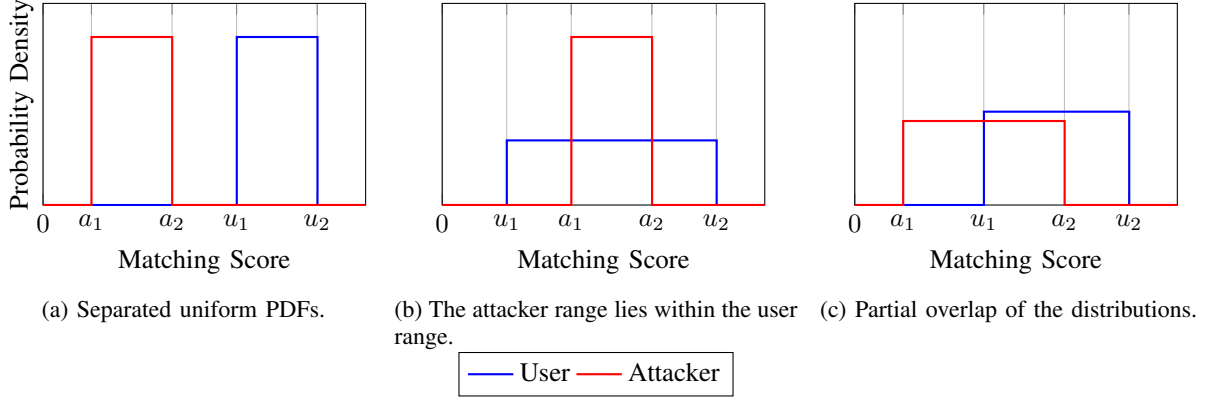


Figure 2: Quantitative illustration of three distinct cases of uniform attacker and user score distributions. In case (1), the distributions are separated. Case (2) illustrate the case where an attacker is fully contained in the user distribution, and (3) illustrate two overlapping configurations.

where the attacker's support lies inside the user's support. Thus, we find the global maximum by differentiating the success probability function in this region and comparing the resulting critical point with the boundary values.

We differentiate the success function and set it to zero. We find that at: $t = \frac{a_1 + u_2}{2}$ we get $\frac{d}{dt} P_{\text{success}}(t) = 0$ with $\frac{d}{dt} P_{\text{success}}(t) > 0$, For every $t \in (u_1, \frac{a_1 + u_2}{2})$ and $\frac{d}{dt} P_{\text{success}}(t) < 0$, For every $t \in (\frac{a_1 + u_2}{2}, a_2)$. Thus, the maximum point is at $t_{\text{Max}} = \frac{a_1 + u_2}{2}$ if: $\frac{a_1 + u_2}{2} \in [u_1, a_2]$ in the overlapping case, or $\frac{a_1 + u_2}{2} \in [a_1, a_2]$ in the case where the attacker's support lies inside the user's support.

For both cases, the probability of success at the optimal point if the condition is satisfied is:

$$P_{\text{success}}(t_{\text{opt}}) = \frac{(a_1 - u_2)^2}{4(a_1 - a_2)(u_1 - u_2)}.$$

For the overlapping case, if the condition is not satisfied the maximum point is at the boundary of the interval, which is either $t = u_1$ or $t = a_2$. Thus:

$$P_{\text{success}}(t_{\text{opt}}) = \max \left\{ \frac{u_1 - a_1}{a_2 - a_1}, \frac{u_2 - a_2}{u_2 - u_1} \right\}.$$

For the case where the attacker's support lies inside the user's support, if the condition is not satisfied the maximum point is at the boundary of the interval, which is either $t = a_1$ or $t = a_2$. Thus:

$$P_{\text{success}}(t_{\text{opt}}) = \max \left\{ 0, \frac{u_2 - a_2}{u_2 - u_1} \right\}.$$

Figure 3 illustrates our analysis of the probability of success for a single biometric credential mechanism as a function of threshold t in the overlapping case. The figure shows that in the second interval the success function is monotonically increasing, while in the fourth interval it is monotonically decreasing. Therefore, the optimal operating point is achieved in the interval where FAR and FRR are strictly between zero and one, which in this scenario is where $u_1 \leq t \leq a_2$.

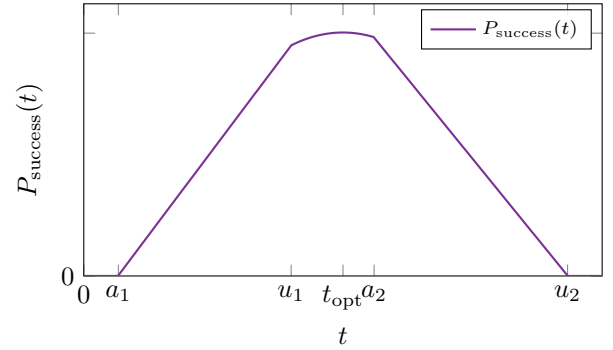


Figure 3: Success probability as a function of the threshold.

We calculate the EER point and show that it is not equal to the optimal point:

$$t_{EER} = \frac{a_1 \cdot u_1 - a_2 \cdot u_2}{a_1 - a_2 + u_1 - u_2} \neq \left\{ a_1, \frac{a_1 + u_2}{2}, u_1, a_2 \right\} = t_{\text{opt}}.$$

The maximum point represents the optimal threshold of the mechanism. From the optimal success probability expression, we can see that when the attacker can produce scores over a wider interval, i.e., when $(a_2 - a_1)$ increases, the mechanism's success probability may significantly decrease. In contrast, a biometric sensor that successfully separates a user and an attacker, by assigning higher scores to a genuine user and lower scores to an attacker, is therefore expected to achieve a higher mechanism's success probability. We calculate the FAR and FRR at the optimal point to find a relation between their values at the optimal point and the EER point:

$$FAR(t_{\text{opt}}) = \frac{2a_2 - a_1 - u_2}{2(a_2 - a_1)} \text{ and}$$

$$FRR(t_{\text{opt}}) = \frac{a_1 + u_2 - 2u_1}{2(u_2 - u_1)}.$$

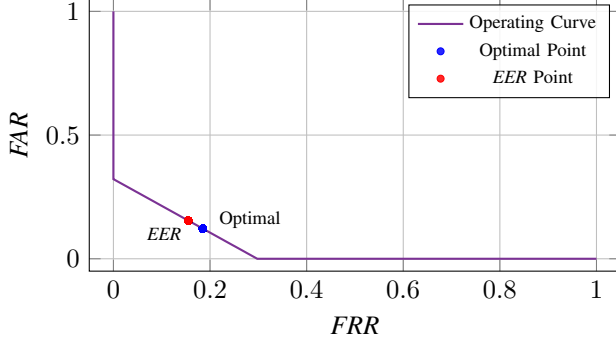


Figure 4: *FAR* vs. *FRR* operating curve with optimal and *EER* points.

Comparing these expressions indicate that when the user and attacker distributions are symmetric, i.e., when the distance between the roots: $a_2 - a_1 = u_2 - u_1$, the *FAR* equals the *FRR*. Figure 4 illustrates our analysis of the *FAR* vs *FRR* operating curve for a single biometric credential mechanism in the overlapping case. We can see that the optimal point differs from the *EER* when the user and attacker distributions are asymmetric. This indicates that as the distributions become closer to symmetric, the distance between the *EER* threshold and the optimal threshold becomes smaller.

4.2. Quadratic Distributed Credential

Our goal is to model credentials whose distributions closely resemble those observed in real measurement data. To this end, we analyze a quadratic probability distribution function, which provides a second order approximation of the biometric credential that can also be seen in real fingerprint measurement data (§7). This choice is also consistent with the findings of Slobodan et al. [12] on Eigenfinger and Eigenpalm feature distributions. Accordingly, we define two probability density functions (PDFs). Let f_U denote the user distribution with roots α_1, α_2 and let f_A denote the attacker distribution with roots β_1, β_2 . Donte by a, b, c and d, e, f the quadratic coefficients of the user and attacker distributions, respectively, hence:

$$f_U(s) = \begin{cases} a \cdot s^2 + b \cdot s + c & \alpha_1 \leq s \leq \alpha_2 \\ 0 & s > \alpha_2 \vee s < \alpha_1 \end{cases}$$

$$f_A(s) = \begin{cases} d \cdot s^2 + e \cdot s + f & \beta_1 \leq s \leq \beta_2 \\ 0 & s > \beta_2 \vee s < \beta_1 \end{cases}$$

To reduce the degrees of freedom of the quadratic distributions while maintaining a symbolic analysis, we express the coefficients directly in terms of their roots. The integral of each PDF over its roots interval is set to 1, which enforces the normalization condition. This condition ensures that the resulting error metrics, *FAR* and *FRR* are bounded to 1. Under this condition, the coefficients become fully determined. Accordingly, we define the coefficients as follows.

$$\alpha_1 \cdot \alpha_2 = \frac{c}{a} \rightarrow a = \frac{c}{\alpha_1 \cdot \alpha_2}$$

$$\alpha_1 + \alpha_2 = -\frac{b}{a} \rightarrow b = -\frac{c \cdot (\alpha_1 + \alpha_2)}{\alpha_1 \cdot \alpha_2}$$

$$\beta_1 \cdot \beta_2 = \frac{f}{d} \rightarrow d = \frac{f}{\beta_1 \cdot \beta_2}$$

$$\beta_1 + \beta_2 = -\frac{e}{d} \rightarrow e = -\frac{f \cdot (\beta_1 + \beta_2)}{\beta_1 \cdot \beta_2}$$

$$\int_{\alpha_1}^{\alpha_2} (a \cdot x^2 + b \cdot x + c) dx = 1 \Rightarrow c = \frac{6 \cdot \alpha_1 \cdot \alpha_2}{(\alpha_1 - \alpha_2)^3}$$

$$\int_{\beta_1}^{\beta_2} (d \cdot x^2 + e \cdot x + f) dx = 1 \Rightarrow f = \frac{6 \cdot \beta_1 \cdot \beta_2}{(\beta_1 - \beta_2)^3}$$

Therefore, we can write the quadratic PDFs as:

$$f_U(s) = \begin{cases} \frac{6}{(\alpha_1 - \alpha_2)^3} s^2 - \frac{6(\alpha_1 + \alpha_2)}{(\alpha_1 - \alpha_2)^3} s + \frac{6\alpha_1\alpha_2}{(\alpha_1 - \alpha_2)^3} & \alpha_1 \leq s \leq \alpha_2 \\ 0, & \text{otherwise} \end{cases}$$

$$f_A(s) = \begin{cases} \frac{6}{(\beta_1 - \beta_2)^3} s^2 - \frac{6(\beta_1 + \beta_2)}{(\beta_1 - \beta_2)^3} s + \frac{6\beta_1\beta_2}{(\beta_1 - \beta_2)^3} & \beta_1 \leq s \leq \beta_2 \\ 0, & \text{otherwise} \end{cases}$$

These distributions establish the error metrics, *FAR* and *FRR*, as a function of the match threshold t . Denote:

$$A(t) = \frac{2(\beta_2^3 - t^3) - 3(\beta_1 + \beta_2)(\beta_2^2 - t^2) + 6\beta_1\beta_2(\beta_2 - t)}{(\beta_1 - \beta_2)^3},$$

$$U(t) = \frac{2(t^3 - \alpha_1^3) - 3(\alpha_1 + \alpha_2)(t^2 - \alpha_1^2) + 6\alpha_1\alpha_2(t - \alpha_1)}{(\alpha_1 - \alpha_2)^3}.$$

$$FAR(t) = \begin{cases} 1, & t \leq \beta_1, \\ A(t), & \beta_1 < t \leq \beta_2, \\ 0, & t > \beta_2, \end{cases}$$

and

$$FRR(t) = \begin{cases} 0, & t \leq \alpha_1, \\ U(t), & \alpha_1 < t \leq \alpha_2, \\ 1, & t > \alpha_2. \end{cases}$$

Similarly to the analysis of the uniform distribution, the x axis represents the matching score s . Genuine users are expected to produce higher matching scores than attacker, thus we analyze three different cases:

(1) $\beta_2 < \alpha_1$: The two distributions are separated. Any threshold t chosen within the range $\beta_2 \leq t \leq \alpha_1$ yield zero *FAR* and *FRR*, and therefore achieves maximal mechanism success as shown in Figure 5a.

(2) $\alpha_1 < \beta_1 < \beta_2 < \alpha_2$: The attacker distribution bounds lies between the range of the user distribution bounds as shown in Figure 5b. The mechanism's probability of success reduces to the safe probability of the biometric credential, which is:

$$P_{\text{safe}} = (1 - FAR(t)) \cdot (1 - FRR(t))$$

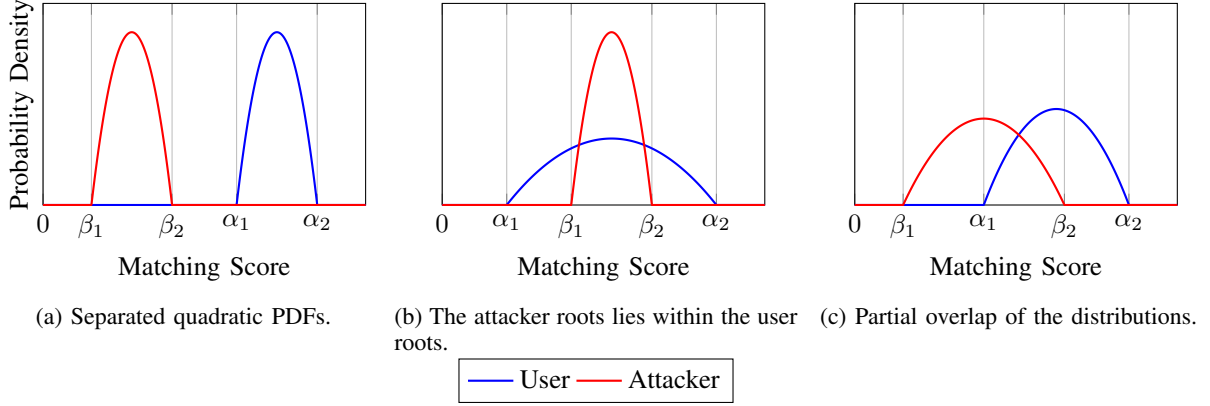


Figure 5: Illustration of three distinct cases of quadratic attacker and user score distributions. In case (1), the distributions are separated. Case (2) illustrates the case where the attacker distribution is fully contained within the user distribution, and case (3) illustrates partial overlap between the two distributions.

By setting the $FAR(t)$ and $FRR(t)$, we can calculate the probability of success. Denote:

$$G(t) = -\frac{(t - \alpha_2)^2(t - \beta_1)^2(2t - 3\alpha_1 + \alpha_2)(2t + \beta_1 - 3\beta_2)}{(\alpha_1 - \alpha_2)^3(\beta_1 - \beta_2)^3}.$$

$$P_{\text{success}}(t) = \begin{cases} 0, & t < \beta_1 \vee t > \alpha_2, \\ G(t), & \beta_1 \leq t \leq \beta_2, \\ -\frac{(t - \alpha_2)^2(2t - 3\alpha_1 + \alpha_2)}{(\alpha_1 - \alpha_2)^3}, & \beta_2 \leq t \leq \alpha_2. \end{cases}$$

(3) $\beta_1 \leq \alpha_1 \leq \beta_2 \leq \alpha_2$: The user and attacker distributions partially overlap like shown in figure 5c. We maintain same approach as the second case and calculate the probability of success. By setting the $FAR(t)$ and $FRR(t)$ into safe probability, the probability of success is:

$$P_{\text{success}}(t) = \begin{cases} 0, & t < \beta_1 \vee t > \alpha_2 \\ \frac{(t - \beta_1)^2(2t + \beta_1 - 3\beta_2)}{(\beta_1 - \beta_2)^3}, & \beta_1 \leq t \leq \alpha_1, \\ G(t), & \alpha_1 \leq t \leq \beta_2, \\ -\frac{(t - \alpha_2)^2(2t - 3\alpha_1 + \alpha_2)}{(\alpha_1 - \alpha_2)^3}, & \beta_2 \leq t \leq \alpha_2. \end{cases}$$

The second and third intervals in the second case are identical to the third and fourth intervals in the third case and therefore, we analyze them together. We differentiate $P_{\text{success}}(t)$ and compute its critical points in every interval. By comparing the stationary values obtained in all valid regions, we identify the global maximum, which corresponds to the optimal threshold t_{opt} .

Before writing the piecewise derivative, we define the polynomial coefficients and the following function:

$$q = 4,$$

$$m = -(5\alpha_1 + \alpha_2 + \beta_1 + 5\beta_2),$$

$$n = \alpha_1(3\alpha_2 + \beta_1 + 6\beta_2) - \alpha_2^2 + \alpha_2\beta_2 - \beta_1^2 + 3\beta_1\beta_2,$$

$$\ell = \alpha_1(\beta_1^2 - 3\beta_2(\alpha_2 + \beta_1)) + \alpha_2^2\beta_2.$$

$$R(t) = -\frac{6(t - \alpha_2)(t - \beta_1)(qt^3 + mt^2 + nt + \ell)}{(\alpha_1 - \alpha_2)^3(\beta_1 - \beta_2)^3}.$$

With these coefficients and function $R(t)$, the derivative of $P_{\text{success}}(t)$ is given by the following piecewise expression:

$$\frac{dP_{\text{success}}}{dt} = \begin{cases} 0, & t < \beta_1 \vee t > \alpha_2, \\ \frac{6(t - \beta_1)(t - \beta_2)}{(\beta_1 - \beta_2)^3}, & \beta_1 \leq t \leq \alpha_1, \\ R(t), & \alpha_1 \leq t \leq \beta_2, \\ -\frac{6(t - \alpha_1)(t - \alpha_2)}{(\alpha_1 - \alpha_2)^3}, & \beta_2 \leq t \leq \alpha_2. \end{cases}$$

In the second interval, the derivative has a single critical point at $t = \beta_1$. Since the derivative is strictly positive for all $t > \beta_1$, this point corresponds to a local minimum rather than a maximum. Similarly, in the fourth interval, the only critical point occurs at $t = \alpha_2$, and the derivative is strictly negative for all $t < \alpha_2$. Therefore, this interval also contains no maximum either.

We therefore turn to the third interval. In this region, the derivative is the product of a cubic polynomial and two linear factors. The linear factors vanish at $t = \beta_1$ and $t = \alpha_2$, both of which lie outside the third interval. Consequently, the existence of critical points within this interval depends solely on the cubic term. Solving this cubic, reveals exactly one real root with the remaining two roots being complex. The closed form of the cubic solution can not be simplified further more than a complex expression. Thus, we denote the solution as t_{cubic} .

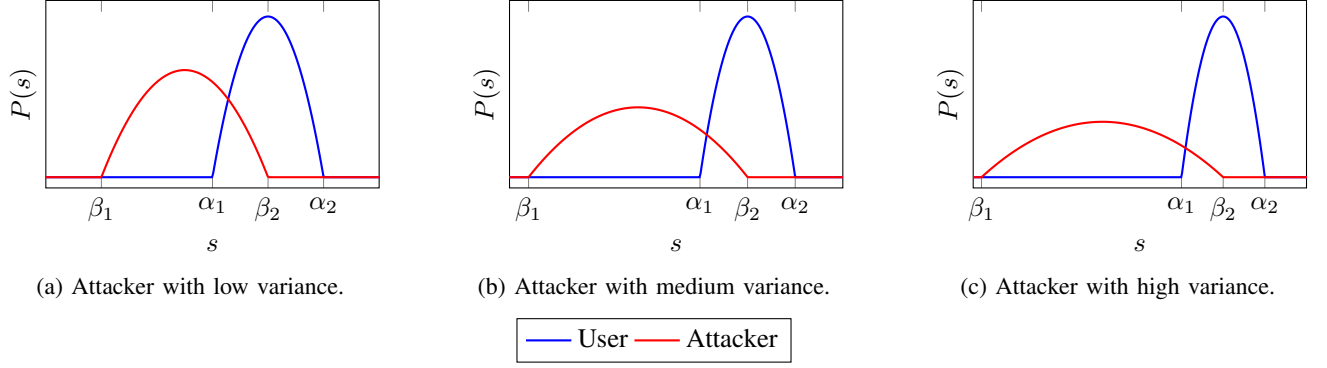


Figure 6: Quadratic user and attacker distributions.

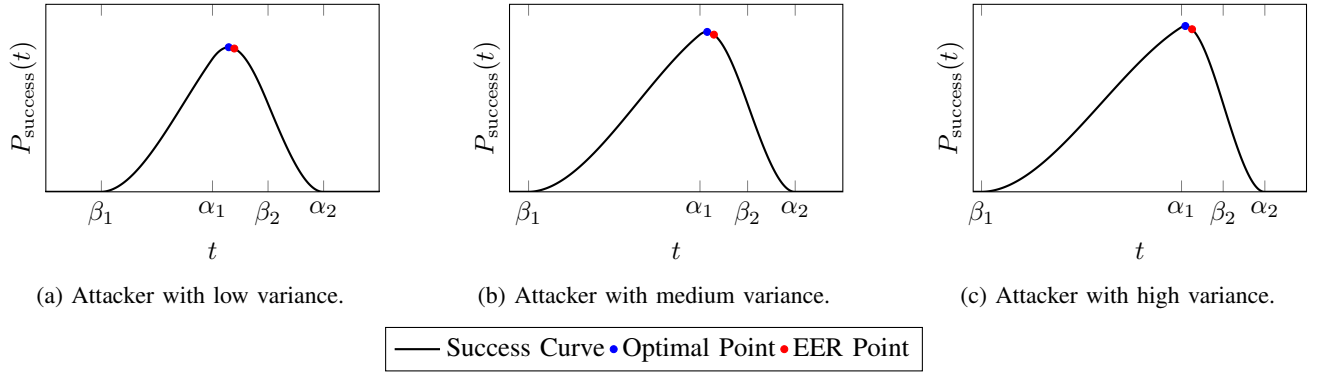


Figure 7: Success probability for three quadratic attacker distributions.

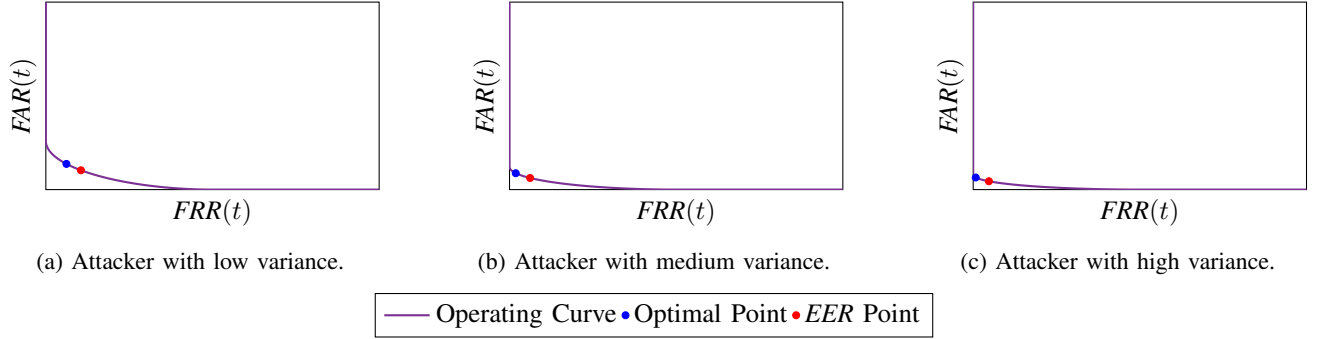


Figure 8: FAR vs. FRR operating curves for different attacker distributions.

By examining the sign of the derivative on both sides of the critical point t_{cubic} , we verify that it constitutes a global maximum of $P_{\text{safe}}(t)$. We denote this maximizing point by t_{opt} .

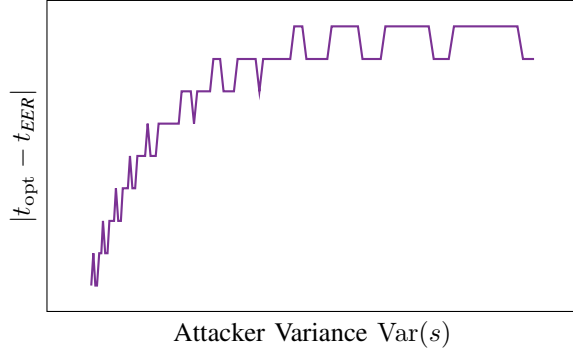
To demonstrate that this optimal threshold differs from the Equal Error Rate (EER) threshold, we computed the EER point separately as follows:

$$t_{EER} = \frac{\alpha_1 \beta_1 - \alpha_2 \beta_2}{\alpha_1 - \alpha_2 + \beta_1 - \beta_2} \neq t_{\text{opt}}$$

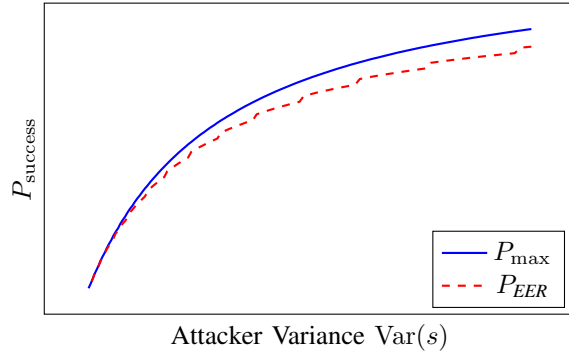
We illustrate several user and attacker distributions and present their corresponding probability of success curves, expressed as functions of the threshold t in Figure 6

and Figure 7 respectively. Comparing each case with its corresponding success probability curve, we observe that the optimal threshold t_{opt} becomes more distinct from the EER threshold t_{EER} as the variance of the attacker distribution increases. We can also see that the gap between the success probability at the optimal point and the success probability at the EER point widens as the attacker's variance increases. This indicates that operating at the optimal threshold can yield a significantly higher success probability compared to operating at the EER threshold, especially when the attacker distribution has a higher variance.

Moreover, we illustrate the FAR - FRR curves for each pair of user and attacker distributions, and mark both the



(a) Threshold gap.



(b) Success at the optimal and *EER* thresholds.

Figure 9: Comparison between the optimal threshold and the *EER* threshold as a function of the attacker's variance.

optimal point and the *EER* point, as shown in Figure 8. We can deduce that as the attacker's variance increases, the optimal point shifts towards α_1 reducing the *FRR* to zero, while the *EER* point remains fixed trying to balance between the *FAR* and *FRR*. This shift indicates that the optimal operating point prioritizes minimizing the false rejection rate, rather than balancing both error rates.

To quantify the effect of the variance on the deviation between the optimal operating point t_{opt} and *EER* point t_{EER} , we numerically computed this gap as a function of the attacker's variance. Decreasing the variance of the attacker achieved by reducing the distance between its roots, corresponds to reduced separability between the two distributions. The simulations reveal that as the variance increases, the gap $|t_{\text{opt}} - t_{\text{EER}}|$ grows accordingly, until the operating threshold moves into a regime where it effectively minimizes only one of the error rates, either *FAR* or *FRR*, driving the other to zero. This phenomenon is shown in Figure 9a. The simulation also indicate that higher attacker variance enlarges the gap between the success probability at the optimal point and the success probability at the *EER* point as show in Figure 9b. This finding highlights the potential success advantage gained by operating at t_{opt} rather than at the *EER* point when an attacker has the ability to produce samples with higher scores values.

5. Two Credentials

For a mechanism M comprises two independently and identically distributed credentials, derived from either the same or different PDF's, we defined the integrated system success as the probability of users successfully defend their assets under two scenarios:

a. *AND*: The user has full availability to all credentials, while the attacker has limited or no availability. For successful authentication, we need both credentials to be either in safe state, or one in safe state and the other in leak state. If one of the credentials is either loss or theft, the owner cannot satisfy the mechanism and thus the mechanism fails.

$$P_{\text{Success}}(M^{\text{AND}}) = P_{\text{safe}}^1 \cdot P_{\text{safe}}^2 + P_{\text{safe}}^1 \cdot P_{\text{leak}}^2 + P_{\text{safe}}^2 \cdot P_{\text{leak}}^1 \quad (2)$$

b. *OR*: The user has availability to at least one credential, while the attacker has no availability neither of them. For success, we need both credential to be either in safe state, or one in safe state and the other in loss state. If one of the credentials is either leak or theft, the attacker can satisfy the mechanism and thus the mechanism fails.

$$P_{\text{Success}}(M^{\text{OR}}) = P_{\text{safe}}^1 \cdot P_{\text{safe}}^2 + P_{\text{safe}}^1 \cdot P_{\text{loss}}^2 + P_{\text{safe}}^2 \cdot P_{\text{loss}}^1 \quad (3)$$

For each mechanism type, we analyze the case of two independently and identically distributed credentials in 2 cases:

- A mechanism comprised of 2 biometric credentials.
- A mechanism comprised of 1 atomic credential and 1 biometric credential.

We derive the optimal operating point for both *AND* and *OR* mechanisms by maximizing the respective success probabilities as function of the thresholds of each credential.

5.1. 2 Biometric Credentials

To analyze the case of a mechanism M comprised of two biometric credentials, we conduct a numerical study over several probability distribution functions. For each distribution pair, we derive the optimal operating point for both *AND* and *OR* mechanisms by maximizing the respective success probabilities with respect to the thresholds of each credential, using a LBFGS optimization algorithm. We additionally compare the optimal operating point to the *EER* point, thereby characterizing the conditions under which the *AND* or *OR* mechanisms yields superior performance. We started by analyzing the case of Gaussian distributions. We defined the user and attacker distributions as $P_U(s) = \mathcal{N}(\mu_u, \sigma_u^2)$ and $P_A(s) = \mathcal{N}(\mu_a, \sigma_a^2)$ respectively. We set for the first credential $\mu_u = 0.8$, $\sigma_u = 0.12$, $\mu_a = 0.3$ and $\sigma_a = 1$ and for the second credential $\mu_u = 1.0$, $\sigma_u = 0.2$, $\mu_a = 0.4$ and $\sigma_a = 0.18$ as shown in Figure 10.

The optimal operating point for both mechanisms' types differ from the *EER* point, yielding higher success probabilities than the *EER* points. Moreover, we find that the optimal points of both *AND* and *OR* mechanisms are not necessarily the same as the optimal threshold if used

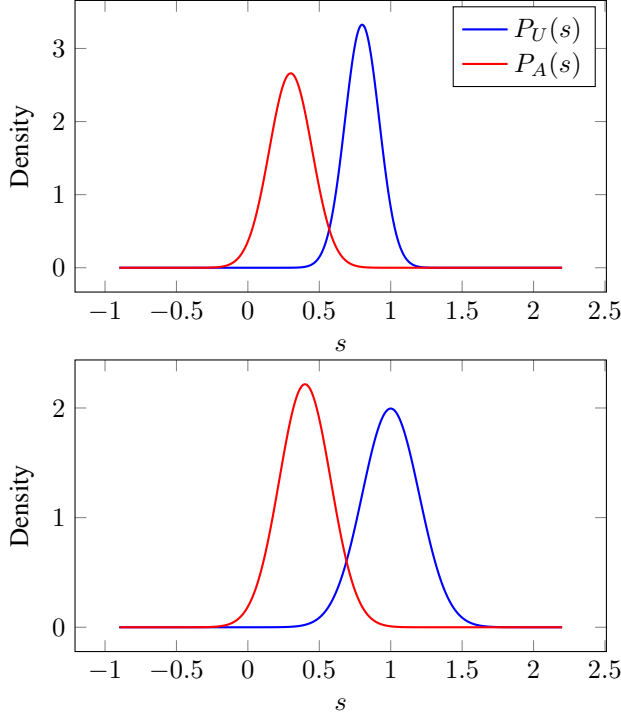


Figure 10: Gaussian user and attacker distributions.

individually. We further find that such integration of a biometric credential to a mechanism already comprised of a biometric credential, improves the mechanism success. Hence, integrating two biometric credentials into a mechanism improves its success over using them individually as shown in Figure 11.

We illustrate the success probability of both *AND* and *OR* mechanisms. We fix a credential to its optimum found out using the LBFGS algorithm and show how the success probability varies as a function of the thresholds of the other credential. The results are shown in Figure 12. In conclusion, all numerical results are summarized in Table 1.

From Table 1, we observe that the optimal operating point yields a higher success probability than the *EER* point by between 5% to 7%.

5.2. 1 Atomic and 1 Biometric Credential

We start by analyzing the case of an *AND* mechanism comprised of 1 atomic credential and 1 biometric credential. Denote the state of credential by s_i , i.e., $i \in (\text{Atomic}, \text{Biometric})$, we define the atomic credential with a constant value of each probabilities' states as: $P_{safe}^1, P_{loss}^1, P_{leak}^1, P_{theft}^1$ corresponding respectively to each one of the states: safe, loss, leak and theft. The biometric credential is defined by its PDFs for the user and the attacker, P_U and P_A respectively. We model the first order approximation of the biometric credential using uniform distribution between the bounds of u_1, u_2 corresponding to the user distribution and a_1, a_2 corresponds to the attacker,

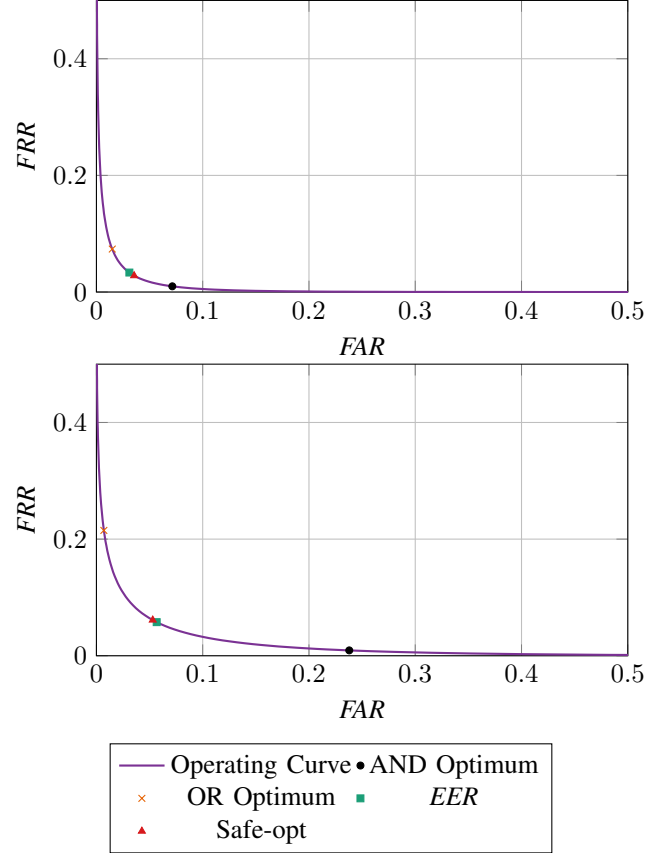


Figure 11: *FAR* vs. *FRR* operating curves for two biometric credentials.

as seen at (§4.1). We derive the probability of the biometric credential as follows:

$$P_{safe}^2 = 1 - FAR(t) - FRR(t) + FAR(t) \cdot FRR(t),$$

$$P_{leak}^2 = FAR(t) (1 - FRR(t)).$$

To analyze the probability of success for the *AND* mechanism, we set the above probabilities into Equation 2. Thus, the probability of success is a function of the threshold t and divided into 5 intervals. Denote:

$$\alpha(t) = \frac{u_2 - t}{u_2 - u_1}, \quad \beta(t) = \frac{a_2 - t}{a_2 - a_1}.$$

$$L(t) = \alpha(t) [(P_{safe}^1 + P_{leak}^1) - P_{leak}^1 \cdot \beta(t)].$$

$$P_{success}(t) = \begin{cases} P_{safe}^1, & t \leq a_1 \\ P_{safe}^1 + P_{leak}^1(1 - \beta(t)), & a_1 < t \leq u_1 \\ L(t), & u_1 < t \leq a_2 \\ \alpha(t) \cdot (P_{safe}^1 + P_{leak}^1), & a_2 < t \leq u_2 \\ 0, & t > u_2. \end{cases}$$

We analyze the success probability $P_{success}(t)$ by partitioning the threshold domain into five intervals. Each interval corresponds to a distinct relative positioning of the threshold with respect to the attacker and user distributions.

Distribution	$P_{\text{AND}}(t_{\text{EER}})$	$P_{\text{OR}}(t_{\text{EER}})$	$P_{\text{AND}}(t_1^{\text{safe}})$	$P_{\text{OR}}(t_2^{\text{safe}})$	$P_{\text{AND}}(t_{\text{opt}})$	$P_{\text{OR}}(t_{\text{opt}})$
Gaussian	0.910	0.911	0.911	0.911	0.964	0.962
Uniform	0.955	0.955	0.916	0.999	1.000	1.000
Parabolic	0.920	0.921	0.919	0.922	0.987	0.985

TABLE 1: Mechanism success under Gaussian, Uniform and Parabolic score distributions at three reference operating points: EER-pair $(t_1^{\text{EER}}, t_2^{\text{EER}})$, individual optimum pair $(t_1^{\text{safe}}, t_2^{\text{safe}})$, and joint optima (AND at $(t_1^*, t_2^*)_{\text{AND}}$ and OR at $(t_1^*, t_2^*)_{\text{OR}}$)

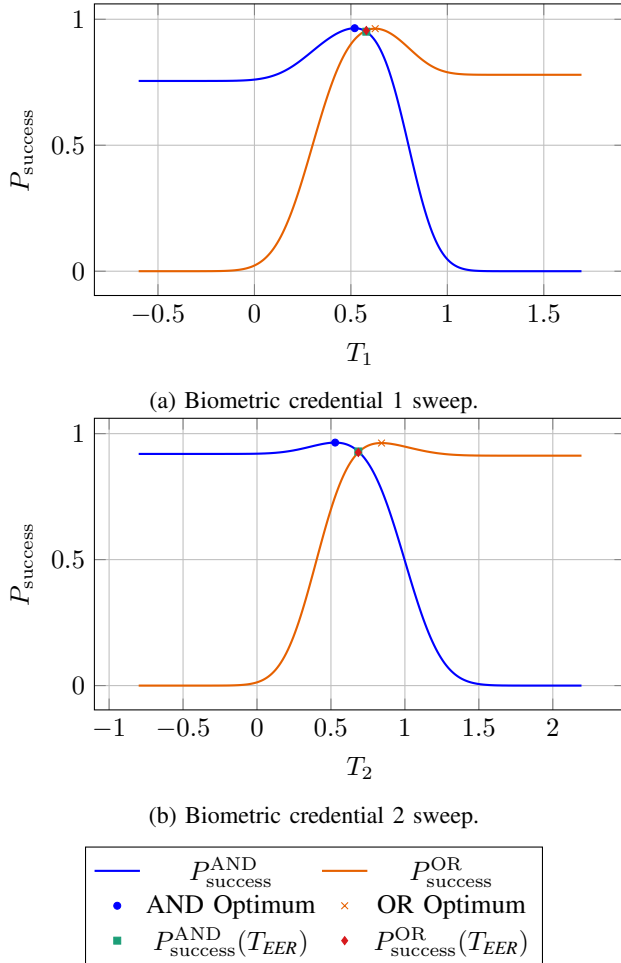


Figure 12: Success probability as a function of the threshold for two biometric credentials.

The first interval, $t \leq a_1$, corresponds to the case in which the threshold is lower than the minimum value of the attacker distribution. In this regime, the FAR is equal to 1, while the FRR is equal to 0. Consequently, the success probability of the biometric credential is zero, and the success probability converges to P_{safe}^1 . Since P_{safe}^1 is constant in this interval, no interior maximum exists.

The second interval, $a_1 < t \leq u_1$, corresponds to the case in which the threshold lies between the minimum value of the attacker distribution and the minimum value of the

user distribution. In this region, the FAR decreases linearly as a function of t , while the FRR remains equal to 0. As a result, the probabilities of the safe state and the leak state of the biometric credential are $1 - FAR(t)$ and $FAR(t)$, respectively. In this interval, $P_{\text{success}}(t)$ is a strictly increasing function, as its derivative is a constant positive value, and therefore the maximum is attained at the boundary point: $t_{\text{opt}2} = u_1$.

The third interval, $u_1 < t \leq a_2$, corresponds to the case in which the threshold lies between the minimum value of the user distribution and the maximum value of the attacker distribution. In this region, both FAR and FRR vary linearly with t , resulting in a success probability that is quadratic as a function of t . Differentiating $P_{\text{success}}(t)$ yields a single critical point inside the interval,

$$t_{\text{opt}3} = \frac{u_2 + a_2 - (a_2 - a_1) \frac{P_{\text{safe}}^1 + P_{\text{leak}}^1}{P_{\text{leak}}^1}}{2}$$

Since this expression is valid only for the interval $u_1 < t \leq a_2$, the effective maximizer within this interval is obtained by clipping the optimum to the interval boundaries, as follows:

$$t_{\text{opt},3}^{(\text{interval})} = \min(a_2, \max(u_1, t_{\text{opt},3})).$$

In particular, if $t_{\text{opt},3} < u_1$, the maximum over this interval is attained at $t = u_1$, whereas if $t_{\text{opt},3} > a_2$, the maximum is attained at $t = a_2$. In all configurations consisting of one atomic and such biometric credential, ingertating the mechanism with the biometric credential monotonically enhances its success probability. Evaluating the success probability at the user threshold $t = u_1$ yields

$$P_{\text{success}}(u_1) = P_{\text{safe}}^1 + P_{\text{leak}}^1 \left(1 - \frac{a_2 - u_1}{a_2 - a_1}\right).$$

Since $u_1 > a_1$, it follows that

$$P_{\text{success}}(u_1) > P_{\text{safe}}^1,$$

Demonstrating that the integration of the biometric credential strictly improves the success of the mechanism compared to using the atomic credential alone.

By examining the optimal point in this interval under AND mechanism, we can deduce that as the variance of the attacker distribution increases, the optimal threshold $t_{\text{opt}3}$ shifts toward the lower boundry of the interval, u_1 . Moreover, for an atomic credential with high safe values P_{safe}^1 , and low leak values P_{leak}^1 , the optimal threshold shifts

toward the lower boundary as well. Thus, in such case, the success of the mechanism becomes mainly dependent on the atomic credential. Conversely, an atomic credential with high leak values P_{leak}^1 and low safe values P_{safe}^1 causes the optimal threshold to shift toward the upper boundary of the interval, a_2 . In this scenario, the success of the mechanism relies more heavily on the biometric credential.

The fourth interval, $a_2 < t \leq u_2$, corresponds to the case in which the threshold exceeds the maximum value of the attacker distribution. In this regime, the FAR is equal to 0, while the FRR increases linearly with t . Consequently, $P_{\text{success}}(t)$ is a strictly decreasing function in this interval, and the maximum is attained at the boundary point

$$t_{\text{opt4}} = a_2.$$

Finally, the fifth interval, $t > u_2$, corresponds to the case in which the threshold exceeds the maximum value of the user and the attacker distributions, leaving the success probability equal to 0.

To determine the global optimal operating point, we compare all stationary points and boundary maxima obtained in the valid intervals. The optimal threshold is achieved in the third interval, where the attacker and user distributions overlap and the success function attains an interior maximum. Substituting $t_{\text{opt}} = t_{\text{opt3}}$ into $P_{\text{success}}(t)$ yields

$$P_{\text{success}}(t_{\text{opt}}) = \frac{(-a_1(P_{\text{safe}}^1 + P_{\text{leak}}^1) + a_2P_{\text{safe}}^1 + P_{\text{leak}}^1 u_2)^2}{4P_{\text{leak}}^1(a_2 - a_1)(u_2 - u_1)}.$$

We have done similar analysis for the *OR* mechanism. We defined the credentials' probabilities of the mechanism as follows:

$$\begin{aligned} P_{\text{safe1}} &= 1 - FAR(t) - FRR(t) + FAR(t) \cdot FRR(t), \\ P_{\text{safe2}} &= k_1, \\ P_{\text{loss1}} &= FRR(t) (1 - FAR(t)), \\ P_{\text{loss2}} &= k_2. \end{aligned}$$

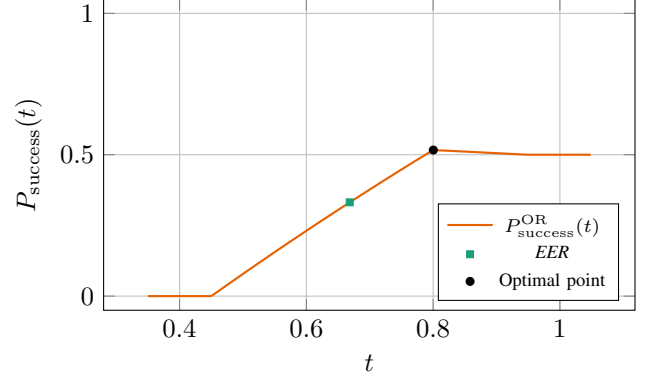
Therefore, the probability of success for the *OR* mechanism is defined as:

$$P_{\text{success}}(t) = \begin{cases} 0, & t \leq a_1 \\ \frac{t - a_1}{a_2 - a_1} (k_1 + k_2), & a_1 < t \leq u_1 \\ \frac{t - a_1}{a_2 - a_1} \left[k_1 + k_2 - k_2 \frac{t - u_1}{u_2 - u_1} \right], & u_1 < t \leq a_2 \\ k_1 + k_2 - k_2 \frac{t - u_1}{u_2 - u_1}, & a_2 < t \leq u_2 \\ k_1, & t > u_2. \end{cases}$$

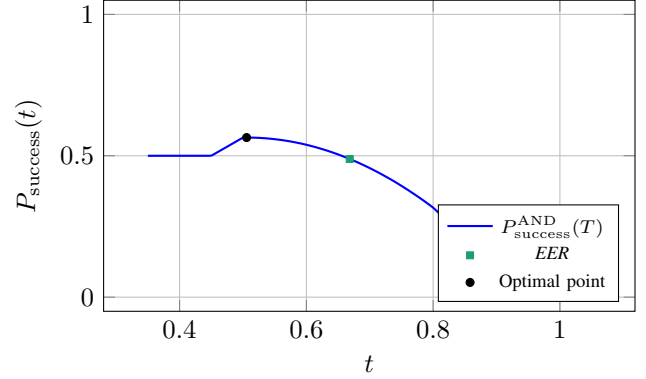
By differentiating every interval and identifying the critical points, we found that the optimal threshold that leads to the maximum probability of success lies in the third interval where

$u_1 < t \leq a_2$ and is given by:

$$t_{\text{opt}} = \frac{a_1 k_2 + u_2(k_1 + k_2) - k_1 u_1}{2k_2}$$



(a) OR mechanism.



(b) AND mechanism.

Figure 13: Success probability as a function of the threshold for OR and AND mechanisms composed of an atomic and a biometric credential.

And the maximum probability of success at this point is:

$$P_{\text{success}}(t_{\text{opt}}) = \frac{(a_1 k_2 - u_2(k_1 + k_2) + k_1 u_1)^2}{4k_2(a_2 - a_1)(u_2 - u_1)}$$

For both mechanisms, we observe that the optimal operating point t_{opt} differs from the equal error rate (EER) point t_{EER} . As the EER depends on the PDF's type solely, we calculate the EER point and show both mechanism's

$$t_{EER} = \frac{a_1 u_1 - a_2 u_2}{a_1 - a_2 + u_1 - u_2}$$

6. Characterizing Success Improvement

We will now characterize the conditions under which integrating a biometric credential into a mechanism composed of an atomic credential can enhance its success. First, we show that for any mechanism whose atomic credential characterized by non-zero probabilities P_{safe} and P_{theft} only, such integration cannot improve mechanism's success (§6.1). Next, we present a necessary and sufficient condition establishing that, for any mechanism composed of an atomic credential with non zero probabilities P_{safe} , P_{leak} or P_{loss} ,

integrating a biometric credential improves mechanism's success for both finite and infinite distributions (§6.2). Finally, we derive a more common criterion that provides a sufficient condition under which a biometric credential improves the mechanism's success composed of an atomic credential with non zero probabilities P_{safe} , P_{leak} or P_{loss} (§6.3).

6.1. No Success Gain in the Safe/Theft Case

Proposition 1. *For any mechanism M composed of an atomic credential with non-zero probabilities P_{safe} and P_{theft} only, integrating the mechanism with a biometric credential does not enhance its mechanism's success.*

Proof. Consider a mechanism M composed of an atomic credential with non-zero probabilities P_{safe} and P_{theft} only. In this case, integrating the mechanism with a biometric credential lead to success probability described by Equation 2 and Equation 3. Since $P_{\text{leak}} = P_{\text{loss}} = 0$, the success probability in both mechanism cases reduces to:

$$P_{\text{success}}(t) = P_{\text{safe}}^1 \cdot P_{\text{safe}}^2(t)$$

Since $P_{\text{safe}}^2(t) \leq 1$ for every value of t , it follows that $P_{\text{success}}(t) \leq P_{\text{safe}}^1$, and thus integrating a biometric credential does not enhance the success of the mechanism. $\square$

6.2. Sufficient And Necessary Condition for Integrating Biometric Credentials

First, we define the notion of *asymptotic separability* of a biometric credential. Then, we will show that its a necessary and sufficient condition for enhancing the mechanism's success for both finite and infinite distributions.

Definition 6. A biometric credential with user distribution function $P_u(s)$ and attacker distribution function $P_a(s)$ is said to be *asymptotically separable* if

$$\inf_t \min \left\{ \frac{FAR(t)}{1 - FRR(t)}, \frac{FRR(t)}{1 - FAR(t)} \right\} = 0,$$

where $FAR(t) = \int_t^\infty P_a(s)ds$ and $FRR(t) = \int_{-\infty}^t P_u(s)ds$.

Proposition 2. *For any mechanism M composed of an atomic credential with non-zero probabilities P_{safe} and P_{leak} or P_{loss} , integrating the mechanism with a biometric credential whose user and attacker distributions are asymptotically separable, enhances the success of the mechanism.*

Proof. The success of a mechanism composed of an atomic credential with non-zero probabilities P_{safe} and P_{leak} or P_{loss} is given by P_{safe}^1 . A biometric credential is characterized by 4 probability states as function of t : $P_{\text{safe}}^2(t)$, $P_{\text{loss}}^2(t)$, $P_{\text{leak}}^2(t)$, $P_{\text{theft}}^2(t)$. We can describe each one of these probabilities as

a function of False acceptance rate (FAR) and False rejection rate (FRR) as follows:

$$\begin{aligned} P_{\text{safe}}^2(t) &= 1 - FAR(t) - FRR(t) + FAR(t) \cdot FRR(t), \\ P_{\text{loss}}^2(t) &= FRR(t)(1 - FAR(t)), \\ P_{\text{leak}}^2(t) &= FAR(t)(1 - FRR(t)), \\ P_{\text{theft}}^2(t) &= FAR(t) \cdot FRR(t). \end{aligned}$$

The success probability of the integrated mechanism comprised of 2 credentials is described by Equation 2 and Equation 3. Thus, by placing the biometric credential in each configuration we can express the success probability of the integrated mechanism as a function of the operating threshold t :

$$P_{\text{success}}^{\text{AND}}(t) = (1 - FRR(t)) \left(P_{\text{safe}}^1 + P_{\text{leak}}^1 (1 - FAR(t)) \right). \quad (4)$$

$$P_{\text{success}}^{\text{OR}}(t) = (1 - FAR(t)) \left(P_{\text{safe}}^1 + P_{\text{loss}}^1 (1 - FRR(t)) \right). \quad (5)$$

We need to show that there exists an operating threshold t such that either $P_{\text{success}}^{\text{AND}}(t) > P_{\text{safe}}^1$ or $P_{\text{success}}^{\text{OR}}(t) > P_{\text{safe}}^1$. By algebraic manipulation of Equation 4 we can demand $P_{\text{success}}^{\text{AND}}(t) > P_{\text{safe}}^1$ if and only if:

$$\frac{FRR(t)}{1 - FAR(t)} < \frac{P_{\text{leak}}^1}{P_{\text{safe}}^1 + P_{\text{leak}}^1 (1 - FAR(t))}$$

Since $P_{\text{leak}}^1 \cdot (1 - FAR(t)) \leq P_{\text{leak}}^1$, a sufficient condition for the above inequality is:

$$\frac{FRR(t)}{1 - FAR(t)} < \frac{P_{\text{leak}}^1}{P_{\text{safe}}^1 + P_{\text{leak}}^1} \quad (6)$$

Similarly, we can deduce that $P_{\text{success}}^{\text{OR}}(t) > P_{\text{safe}}^1$ if and only if:

$$\frac{FAR(t)}{1 - FRR(t)} < \frac{P_{\text{loss}}^1}{P_{\text{safe}}^1 + P_{\text{loss}}^1 (1 - FRR(t))}$$

Since $P_{\text{loss}}^1 \cdot (1 - FRR(t)) \leq P_{\text{loss}}^1$, a sufficient condition for the above inequality is:

$$\frac{FAR(t)}{1 - FRR(t)} < \frac{P_{\text{loss}}^1}{P_{\text{safe}}^1 + P_{\text{loss}}^1} \quad (7)$$

By the definition of asymptotic separability, there exists an operating threshold t such that:

$$\inf_t \min \left\{ \frac{FAR(t)}{1 - FRR(t)}, \frac{FRR(t)}{1 - FAR(t)} \right\} = 0.$$

Equivalently, $\forall \varepsilon > 0, \exists t$ such that either $\frac{FAR(t)}{1 - FRR(t)} < \varepsilon$ or $\frac{FRR(t)}{1 - FAR(t)} < \varepsilon$. Thus, we can choose ε that equals the right hand side of either Equation 6 or Equation 7 and by definition we know that a t satisfies the inequality exists. More precisely, if

$$\inf_t \frac{FAR(t)}{1 - FRR(t)} = 0,$$

then the OR mechanism composition is preferable, whereas if

$$\inf_t \frac{FRR(t)}{1 - FAR} = 0,$$

then the AND mechanism composition is preferable. Therefore, integrating an asymptotically separable biometric credential enhances the success of the mechanism. $\square$

Proposition 3. Suppose there $\exists t$ such that the integrated mechanism M composed of a biometric credential and an atomic credential with non zero probabilities P_{safe} and P_{leak} or P_{loss} achieves a strictly higher success probability than an atomic credential mechanism, i.e.,

$$P_{\text{success}}^{\text{integrated}}(t) > P_{\text{success}}^{\text{atomic}}.$$

Then necessarily, the biometric credential is asymptotically separable.

Proof. We prove by contradiction. Assume for the sake of contradiction, that the biometric credential is not asymptotically separable. Then $\exists \varepsilon > 0$ such that $\forall t$ both $\frac{FAR(t)}{1 - FRR(t)} \geq \varepsilon$ and $\frac{FRR(t)}{1 - FAR} \geq \varepsilon$. Since it's given that the integrated mechanism M achieves a strictly higher success probability than an atomic credential mechanism, then there exists a threshold t such that either $P_{\text{success}}^{\text{AND}}(t) > P_{\text{safe}}^1$ or $P_{\text{success}}^{\text{OR}}(t) > P_{\text{safe}}^1$. Thus, either Equation 6 or Equation 7 holds for every value of $P_{\text{safe}}^1, P_{\text{leak}}^1, P_{\text{loss}}^1$ in the range of $(0, 1)$. In particular, we can choose $P_{\text{safe}}^1, P_{\text{leak}}^1, P_{\text{loss}}^1$ such that either $\frac{P_{\text{leak}}^1}{P_{\text{safe}}^1 + P_{\text{leak}}^1} < \varepsilon$ or $\frac{P_{\text{loss}}^1}{P_{\text{safe}}^1 + P_{\text{loss}}^1} < \varepsilon$ since the equation holds $\forall \varepsilon > 0$, but both $\frac{FRR(t)}{1 - FRR(t)} \geq \varepsilon$ and $\frac{FAR(t)}{1 - FAR} \geq \varepsilon$ derive by the sake of contradiction assumption. Thus, we can deduce that both $\frac{FRR(t)}{1 - FAR(t)} \geq \frac{P_{\text{leak}}^1}{P_{\text{safe}}^1 + P_{\text{leak}}^1}$ and $\frac{FAR(t)}{1 - FRR(t)} \geq \frac{P_{\text{loss}}^1}{P_{\text{safe}}^1 + P_{\text{loss}}^1}$, which contradicts both Equation 6 and Equation 7 under the strictly higher success probability assumption. Therefore, the biometric credential must be asymptotically separable. $\square$

We show that *asymptotic separability* of the user and attacker distributions is a sufficient and necessary condition for enhancing the success of a mechanism composed of an atomic credential with non zero probabilities $P_{\text{safe}}, P_{\text{leak}}$ or P_{loss} by integrating it with a biometric credential.

6.3. Common Sufficient Condition

Definition 7. A biometric credential with user distribution function $P_u(s)$ and attacker distribution function $P_a(s)$ is said to be *asymptotically dominated* if:

$$\forall c > 0 \exists t \text{ s.t. : } \forall s \geq t : p_a(s) \leq c p_u(s)$$

or

$$\forall c > 0 \exists t \text{ s.t. : } \forall s \leq t : p_u(s) \leq c p_a(s)$$

We will show that *asymptotically dominancy* of user and attacker distribution functions is a sufficient condition for enhancing an atomic credential mechanism's success.

Proposition 4. For any mechanism M composed of an atomic credential and success function $P_{\text{success}}^{\text{atomic}}$, integrating the mechanism with an asymptotically dominated biometric credential improves its achievable success. That is,

$$P_{\text{success}}^{\text{integrated}}(t) > P_{\text{success}}^{\text{atomic}}.$$

Proof. An atomic credential is characterized by fixed probability states $P_{\text{safe}}^1, P_{\text{loss}}^1, P_{\text{leak}}^1, P_{\text{theft}}^1$. Thus the success probability of a mechanism composed solely of this atomic credential is constant and given by:

$$P_{\text{success}}^{\text{atomic}} = P_{\text{safe}}^1$$

A biometric credential is characterized by 4 probability states that depend on the operating threshold t : $P_{\text{safe}}^2(t), P_{\text{loss}}^2(t), P_{\text{leak}}^2(t), P_{\text{theft}}^2(t)$. We can describe each one of these probabilities as a function of False acceptance rate (FAR) and False rejection rate (FRR) as follows:

$$P_{\text{safe}}^2(t) = 1 - FAR(t) - FRR(t) + FAR(t) \cdot FRR(t),$$

$$P_{\text{loss}}^2(t) = FRR(t)(1 - FAR(t)),$$

$$P_{\text{leak}}^2(t) = FAR(t)(1 - FRR(t)),$$

$$P_{\text{theft}}^2(t) = FAR(t) \cdot FRR(t).$$

The success probability of the integrated mechanism comprised of 2 credentials is described by Equation 2 and Equation 3. Thus, we can express the success probability of the integrated mechanism as a function of the operating threshold t as shown in Equation 4 and Equation 5. We previously observed that a sufficient condition for improvement of mechanism's success is described by Equation 6 and Equation 7. Thus, we need to show that there exists an operating threshold t such that either:

$$\frac{FRR(t)}{1 - FAR(t)} < \frac{P_{\text{leak}}^1}{P_{\text{safe}}^1 + P_{\text{leak}}^1}$$

or

$$\frac{FAR(t)}{1 - FRR(t)} < \frac{P_{\text{loss}}^1}{P_{\text{safe}}^1 + P_{\text{loss}}^1}$$

We expand left hand side of Equation 6.

$$\frac{FRR(t)}{1 - FAR(t)} = \frac{\int_{-\infty}^t P_u(s) ds}{\int_{-\infty}^t P_a(s) ds}$$

By Definition 7, since the biometric credential is asymptotically dominated, it follows that there exists an operating threshold t such that for every $s \leq t$, $P_u(s) \leq c \cdot P_a(s)$. Therefore:

$$\frac{\int_{-\infty}^t P_u(s) ds}{\int_{-\infty}^t P_a(s) ds} \leq \frac{\int_{-\infty}^t c \cdot P_a(s) ds}{\int_{-\infty}^t P_a(s) ds} = c$$

This is true $\forall c > 0$, thus we can choose c such that satisfies Equation 6.

We expand left hand side of Equation 7.

$$\frac{FAR(t)}{1 - FRR(t)} = \frac{\int_t^\infty P_a(s)ds}{\int_t^\infty P_u(s)ds}$$

By Definition 7, since the biometric credential is asymptotically dominated, it follows that there exists an operating threshold t such that for every $s \geq t$, $P_a(s) \leq c \cdot P_u(s)$. Therefore:

$$\frac{\int_t^\infty P_a(s)ds}{\int_t^\infty P_u(s)ds} \leq \frac{\int_t^\infty c \cdot P_u(s)ds}{\int_t^\infty P_u(s)ds} = c, \forall c > 0$$

We showed that $\forall c > 0$ either $\frac{FAR(t)}{1 - FRR(t)} < c$ or $\frac{FRR(t)}{1 - FAR(t)} < c$. Therefore, integrating an asymptotically dominated biometric credential with an atomic credential improves mechanism success. $\square$

This proof is not trivial, as the integration of another atomic credential to an atomic credential mechanism, does not always enhance its success. For example, if the mechanism is comprised of 1 atomic credential with high safe values and low leak values, adding an atomic credential with low safe values and high leak values would decrease the overall mechanism's success. Thus, Proposition 4 shows that by using an *asymptotically dominated* biometric credential, we can always improve the success of all mechanism's composed of an atomic credential. Moreover, *asymptotically dominancy* condition supports practical scenarios, as we expect that for very low values of score s the attacker distribution dominates the user, vice versa for very high values of s . We supports this claim in with real public biometric data in Section 7.

7. Measurements

We now present experiment measurements results based on fingerprints taken from *FVC2002* database [26] and *FVC2004* database [27]. We used a standard fingerprint verification algorithm *SourceAFIS* created by Vazan [13] for scoring fingerprints out of their images based on minutiae triplets. We chose to use fingerprints, as it is a widely common and used example of a biometric credential in authentication mechanisms. For a test set, we took a template out of *FVC2002* database 1 of user 103 and compare them against the whole database of both *FVC2002* and *FVC2004*. A genuine will be considered the template of user 103 against its other templates while imposter considered as all the other combinations. For scoring methodology, we define the final score as the maximum similarity across all 5 user templates. For example, if the matching scores between a sample and the 5 probes are $[0.1, 0.3, 0.6, 0.9, 0.4]$, then the resulting score is 0.9.

We maintain same approach as Jain et al. [3] and Cappelli et al. [22] and first built histogram of the genuine and imposter combinations as shown in Figure 14. Then, we turn the histogram into a smoothed probability distribution function. We find the False rejection rate and False acceptance

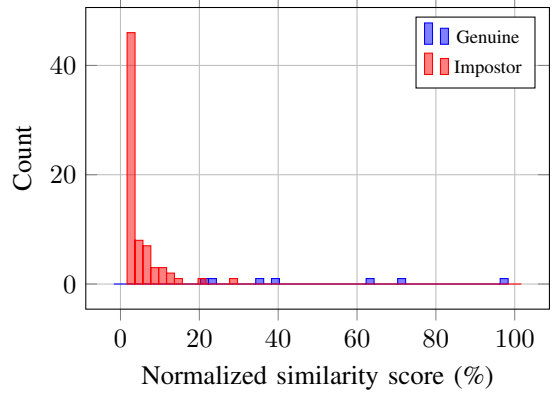


Figure 14: Score histogram for genuine and impostor comparisons.

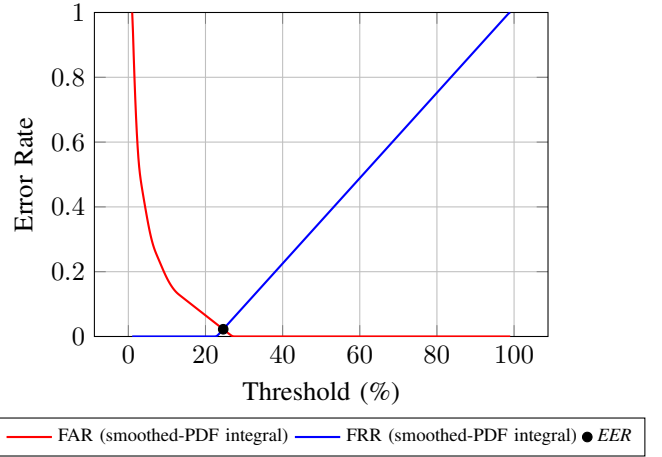
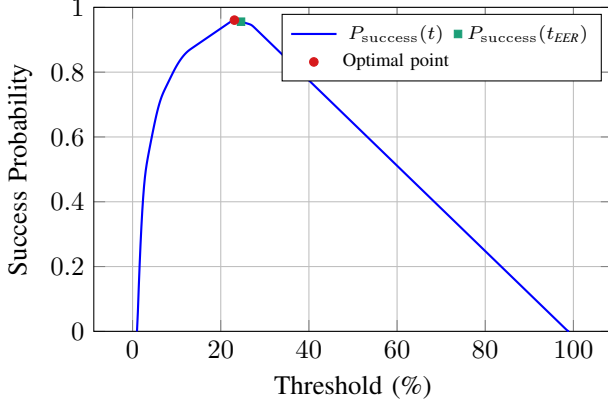


Figure 15: *FAR* and *FRR* as a function of the decision threshold, with the Equal Error Rate (*EER*) point.

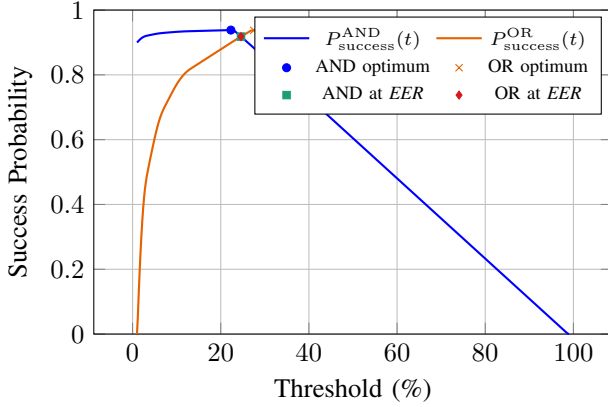
rate and calculate the Equal Error Rate (*EER*) as shown in Figure 15.

The most important outcome from this approach was to achieve similar *EER* of around 2% as achieved from Maio et al. [23], to objectively make a comparison with state-of-the-art algorithms. We calculate the probability success function as given in Equation 1 of a mechanism composed of a biometric credential and show that the optimal operating point is different from the *EER* point and achieves higher success probability as shown in Figure 16. Finally, we illustrate the probability success function of a mechanism composed of an atomic credential with $P_{\text{safe}} = 0.9$ and $P_{\text{leak}} = 0.04$, $P_{\text{loss}} = 0.04$ and integrate it with the biometric credential.

We show that the optimal operating point of the integrated mechanism is different from the *EER* point and achieves higher success probability by around 3% compared to the *EER* point, as the *EER* itself is only 1.8%. In addition, we can observe that the optimal point in both mechanisms achieves higher success probability than both the atomic credential mechanisms and biometric credential mechanism



(a) Single biometric credential mechanism's success.



(b) biometric credential integrated with an atomic credential into AND and OR mechanisms.

Figure 16: Success probability as a function of the threshold. The AND and OR mechanisms achieve success probabilities of 0.93 and 0.94 at the *EER* point, and 0.96 and 0.95 at their respective optimal operating points.

when tested individually as shown in Figure 16. Although there is an improvement in the success probability of the integrated mechanism, we can observe the improvement is not as significant as the one observed in the theoretical analysis. This can be explained by the fact that the *EER* point of the biometric credential is already very low, indicating high separability between a user and an attacker and thus, there is no much room for improvement. In that scenario, we compared several templates of user 103 against templates of different users. However, our main approach in this paper was to increase the mechanism's success in scenarios where an imposter will try to spoof the user templates and authenticate into the user account. Hence, we expect the imposter to use spoofing tools, producing samples resamble with score distribution function that much more similar to the genuine distribution function, which might lead to increased *EER* point.

To support this claim, we used the LivDet2015 database [14], [15], [16], [17], which contains a large-scale

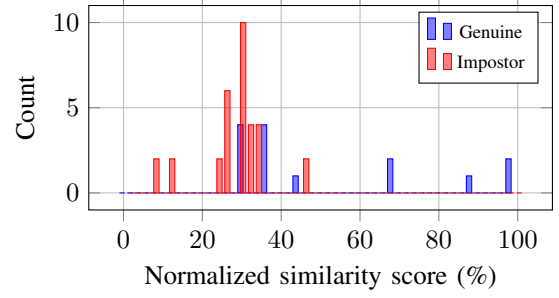


Figure 17: LivDet score histogram using the best score over probe templates.

database of live and spoof fingerprint samples. We follow the same methodology as before and used the SourceAFIS algorithm [13]. As probe templates, we selected six Digital Persona scanner images of user 002_0. Each probe was matched against 14 additional images of the same user to generate genuine scores. For each target image, we define the final score methodology as the maximum similarity across 6 probe templates. For example, if the matching scores between a target image and the six probes are [0.3, 0.5, 0.4, 0.8, 0.9, 0.7], then the resulting score is 0.9. To simulate a spoofing attack, we used fake fingerprint images of the same user (002_0) produced by Gelatine, Woodglue, Latex and Ecoflex and match them against the probe templates to obtain the imposter score distribution. The same maximum-score aggregation was applied in this case as well. We present the resulting histogram in Figure 17. We then compute the *EER* point and plot the *FAR-FRR* curve, as shown in Figure 18. As expected, the *EER* increased significantly to 17.8%. Finally, we evaluated the success probability for both a single biometric credential and for a mechanism composed of an atomic credential and a biometric credential, as shown in Figure 19. We can observe that the optimal operating point reducing the failure probability of an AND mechanism by 60% and OR mechanism by 44% compared to the *EER* point. Moreover, we reduced the failure probability by 50% when integrating into AND mechanism compared to an atomic credential mechanism with $P_{\text{safe}} = 0.8$, $P_{\text{loss}} = 0.04$, $P_{\text{leak}} = 0.15$, $P_{\text{theft}} = 0.01$ probabilities.

8. Conclusion

We formalize the authentication mechanism model consisting, for the first time, of both biometric and atomic credential mechanisms. We analyze the success of each mechanism type and present a new operating point that achieves higher success probability by modelling each credential with four probability states. We show every mechanism comprised of atomic credentials can be enhanced by integrating it with asymptotically separable biometric credential. Our analysis shows significant reduction of the failure probability under spoofing attempts, which became prevalent and easier on biometric systems in today's artificial

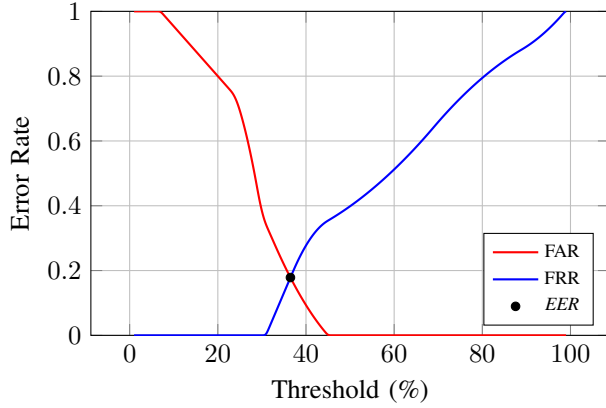
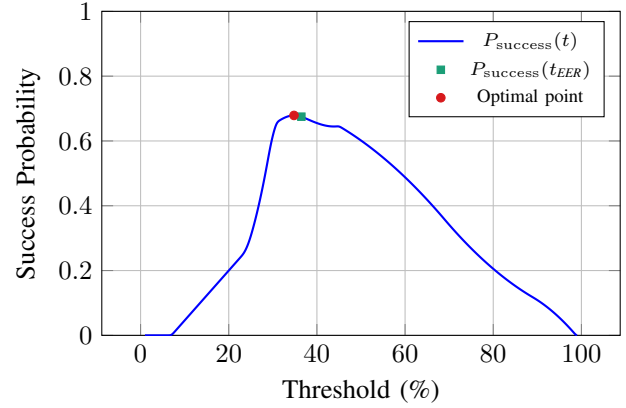


Figure 18: *FAR* and *FRR* as a function of the threshold using the best score over probe templates. The Equal Error Rate (*EER*) occurs at $\approx 17.8\%$.

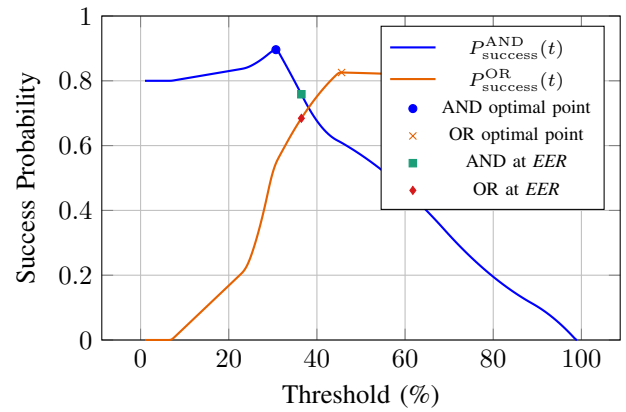
intelligence era. Building new mechanisms, based on the suggested dynamic operating point done (§4) is a promising direction for future work. Our findings have important and impactful implications for the design of secure authentication biometric systems, as we suggest letting the integrators to set the thresholds of the biometric sensors. Harnessing our approach, may affect millions of users around the world to enhance their security of digital and physical assets.

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(a) Single biometric credential.



(b) AND and OR mechanisms.

Figure 19: Success probability as a function of the decision threshold using the best score over probe templates. The AND mechanism achieves a maximal success probability of 0.896, while the OR mechanism achieves 0.826. At the *EER* operating point, the success probabilities are 0.759 and 0.684 for the AND and OR mechanisms, respectively.

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Appendix